21 (a) Show that $\sqrt{800}$ can be written in the form $a\sqrt{2}$ where a is an integer.

$800 = 400 \times 2$

$\sqrt{800} = \sqrt{400} \times \sqrt{2} = 20\sqrt{2}$

Therefore $a = 20$.

2 marks total
Take the largest square factor outside the root.

(2)

(b) Rationalise the denominator of $\dfrac{1+\sqrt{5}}{5-\sqrt{5}}$

Give your answer in its simplest form.

$$\frac{1+\sqrt{5}}{5-\sqrt{5}} \times \frac{5+\sqrt{5}}{5+\sqrt{5}}$$

1 mark
Multiply by the conjugate over itself.

$(5-\sqrt{5})(5+\sqrt{5}) = 25 - 5 = 20$

$(1+\sqrt{5})(5+\sqrt{5}) = 10 + 6\sqrt{5}$

1 mark
Expand both products.

$$\frac{10+6\sqrt{5}}{20} = \frac{5+3\sqrt{5}}{10}$$

1 mark
Simplify by dividing the whole numerator and denominator by 2.

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(3)

(Total for Question 21 is 5 marks)

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